

Peyton Hall

Assignment 4: Populating a Database Design with Data and Querying It

Professor Kazemzadeh

The data model represents a simplified portion of the periodic table. The domain is chemistry, viz. the organization of chemical elements by classification and periodic table position. This design was carried forward from Assignment 3, in which the logical and relational models were constructed using Oracle SQL Developer Data Modeler.

As with the previous assignment, there are three entities: Element, Category, and Period. Element stores the atomic number, element name, symbol, atomic mass, period, group number, and category of each chemical element, along with foreign key references to Category and Period. Category represents the chemical classification of an element, which may be Metal, Nonmetal, Noble Gas, or Metalloid, and contains a unique category_id. Period represents the horizontal rows of the periodic table and contains a unique period_id. Two one-to-many relationships are defined: Category to Element, because each element belongs to exactly one category while a category may contain many elements, and Period to Element, because each element belongs to exactly one period while a period may contain many elements.

The DDL was written by hand in SQL Developer after the original export from Oracle Data Modeler produced UNKNOWN data types, which required manual correction. The tables were created and populated with all 118 confirmed elements of the periodic table using INSERT statements. The five required queries were then written and executed: (1) a SELECT statement with an explicit JOIN across all three tables, (2) an UPDATE statement modifying the atomic mass of Hydrogen, (3) a DELETE statement against the Category table which was prevented by the foreign key constraint on Element, (4) a CREATE VIEW statement producing a view of element names and their categories, and (5) an ALTER TABLE statement adding a discovery_year column to the Element table. All statements ran without error, with the exception of the DELETE, which was prevented by the foreign key constraint as required by the assignment.